APPENDIX PMD — POSITION–MASS DUALITY

The Context-Price of a Definite Location — how *localization* is bought with **mass** as well as with **space**

> *"To be anywhere in particular, a thing must pay twice: once in space, by giving up everywhere; and once in mass, by letting the rest of the universe hold it in place. A 'here' is context that has begun to weigh."*
> — Petar Nikolov, opening the PMD hypothesis, 2026-07-19
>
> *(Figurative framing. The defensible core claim is **not** gravitational weight or a literal Machian mechanism — it is §30's Localization-Leverage: mass sets localization **capacity**, context realizes **access**. The Machian reading is flagged open throughout, §4 / §8 / §30.)*

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Framework: U-Theory v31 · deepens the **Position** ↔ **Space** currency
Status: L3 / L4 SPECULATIVE (Part I) + L1/L2/L3 FORMAL (Part II) — a coherent *interpretive* extension of the canon, **NOT** a derivation of new physics, **NOT** a modification of GR or QFT. **(v1.4 — review-hardened: §17 adapter adopted into the §5 core meter (`P_L = P_D^(1−η)·P_M^η`, η=½ default; `P_L=P_M` retained as the η=1 legacy case); context = channel vs its Fisher content clarified; PMD-2 NR-scope, PMD-4 regularity, PMD-6 rival-aggregator, and raw-observable protocol notes added.)**
Date (this revision): 2026-07-25
Prerequisites: the core triad `Form⇔Time · Position⇔Space · Action⇔Energy`; `APPENDIX_MMT` (Meaning–Matter Transformation), `APPENDIX_DIM` (Dimensionless Meaning), `APPENDIX_ST` / DPR (Dimensional Price Registry), `APPENDIX_QMC` (measurement collapse), `APPENDIX_GEN` (Genesis Law), `APPENDIX_SSS` (the score); for Part II, basic quantum estimation theory (Fisher information, Cramér–Rao, quantum Fisher information) and `APPENDIX_RH` (null-model / rival discipline).

> **How to read this document. PART I (§0–§8)** is the *interpretation*: what the Position↔Mass duality means, its honest scope, and where it is speculative (L3/L4). **PART II (§9–§30)** is the *formal layer*: what can actually be **proved** (identities and information inequalities, L1/L2), what is a **declared modelling choice** (the `[0,1]` meters, L3), and what is **forbidden** without a new observable (the no-go theorem). If

> you want anything quantitative, go to Part II. **Minimal reading paths:** *conceptual reader* — Part I only, stop at §8; *quantitative user* — §12 (the bound), §17 (the adapter), §23 (the no-go), §30 (the core claim).

══════════════════════════════════ PART I — INTERPRETATION ══════════════════════════════════

0. What this appendix IS and IS NOT

This appendix **IS**	This appendix **IS NOT**
A sharpening of the **Position** currency: a location has *two* non-substitutable prices, not one	A new ontology or a new fundamental theory
A reading of the real, textbook link between **mass and localizability** through the F/P/A lens	A derivation from the L1 axioms
Anchored on established physics where it is established (Compton wavelength, inertia)	A claim to overturn or replace GR / QFT
Explicit about which step is rigorous and which is speculative interpretation	A claim of empirical proof of "context = mass"
Marked **L3/L4 speculative** throughout, with honest failure hooks (§8)	A claim that a delocalized state *must* be massless (it need not — §8)

Core thesis (domain-limited). *Position* ↔ *Space* is the canon. PMD adds a **narrower** bridge: **for a massive one-particle state in a chosen rest-frame description**, invariant mass fixes the reduced-Compton scale `ƛ_C = ħ/(mc)` at which a fixed-particle-number localization becomes QFT-sensitive, and it sets the **inertial response to changes in momentum**. U-Theory reads these two roles as a **mass-sensitive sub-meter within the Position pillar** — *one* proxy for localizability, not a universal one. **Massless field excitations (photons) need a separate localization/detection meter and are NOT assigned zero Position merely because their invariant mass vanishes** (§5, §8).

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1. The question (stated cleanly)

The canon says **resistance-against-Form = Time**, **resistance-against-Position = Space**, **resistance-against-Action = Energy**. Position is priced in Space: to occupy a unique location you must **exclude** a region — you give up being everywhere.

But a location has a *second* face the canon leaves implicit:

- For a **massive** particle, a sharp rest-frame **"here"** is bounded below by the reduced-Compton scale `ƛ_C = ħ/(mc)`, so mass **sets the rest-frame localization scale** *(it does not follow that massless quanta cannot be localized at all — §8)*.
- To **change its state of motion**, a system must overcome **inertia** — resistance to a *change of momentum*, `F = dp/dt` (reducing to `F ≈ m·a` only for `v ≪ c`).

So the Position currency splits into two coupled sub-prices — **Space** (the exclusion/extent face) and **Mass** (the context/inertia face). This appendix makes that duality explicit and connects it to the rest of the corpus.

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2. The rigorous anchor — the Compton wavelength (established physics)

This section is **not** speculative. It is textbook relativistic quantum mechanics, and it is the load-bearing core on which the interpretation rests.

The **Compton wavelength** of a particle of mass `m` is:

```
ƛ_C = ħ / (m·c)          (the REDUCED Compton wavelength)
λ_C = h / (m·c) = 2π · ƛ_C   (the ordinary Compton wavelength)
```

It is the **characteristic scale at which a fixed-particle-number, one-particle description becomes unreliable** (QFT particle-production becomes relevant) — **not** an unconditional, Lorentz-invariant minimum width of every possible state. Sketch: to pin a particle to a region of size `Δx` you must inject momenta of order `ħ/Δx` (uncertainty principle), i.e. energies of order `ħc/Δx`. Once `Δx ≲ ƛ_C`, that energy reaches `~m·c²` — enough to **pair-create** — and the very idea of "one particle at a point" dissolves. Consequences (within the one-particle description):

- `m → 0` ⇒ `ƛ_C → ∞` ⇒ **no rest-frame one-particle localization scale.** A massless excitation (the photon) has **no rest frame** and no standard sharp one-particle position operator (there is none for massless spin > ½), so it cannot be pinned as a rest-frame "here." *(Nuance, §8: it CAN still be prepared as a finite wave packet and register **local detection events** — masslessness forbids the rest-frame particle-"here", not every local detection.)*
- **Larger** `m` ⇒ **smaller** `ƛ_C` ⇒ **a finer one-particle localization scale.** Mass **sets** that scale (and the inertial response, §3); PMD reads it as the *mass sub-price* of a definite Position.

> **Rigorous takeaway (physics, not analogy):** mass **sets the finite one-particle, rest-frame localization scale** `ƛ_C` and the inertial response. More mass → a finer rest-frame "here"; zero mass → no rest-frame "here" (local detections still occur). PMD *reads* this scale as the mass sub-price of Position — the physical fact (the scale) is not in dispute, only the reading is.

> **Technical note — reduced scale & localization scope.** Here `ƛ_C = ħ/(mc)` is the **reduced** Compton wavelength (ordinary `λ_C = h/(mc) = 2π·ƛ_C`). The localization claim is about the breakdown of a **stable one-particle, rest-frame** description: confining a *massive* particle to `Δx ≲ ƛ_C` makes particle creation relevant. For *massless* quanta (photons) the point is not that no local detection can occur, but that there is **no rest frame** and **no standard sharp Newton–Wigner one-particle position observable**. PMD reads mass as *enabling a finite rest-frame localization scale*, not as the sole cause of all spatial detection.

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3. The second face — inertia (the price of *changing* a location)

Newton, relativistically safe: `F = dp/dt` with `p = γ·m·v`, reducing to `F ≈ m·a` only for `v ≪ c`. **Inertial mass is resistance to a *change of motion / position*** (acceleration). Placing this beside the canon:

what is being resisted	the price (currency)
Form being changed (endurance in time)	**Time**
occupying a location (exclusion in space)	**Space**
Action being performed (work)	**Energy**
changing its state of motion (acceleration / Δmomentum)	**Mass (inertia)**

> **Note (mass is not a fourth pillar).** This row places **Mass** beside Time/Space/Energy only to display the *parallel structure* of "what is resisted → the price." It does **not** promote mass to a fourth top-level currency. Mass is a **sub-price nested inside the Position pillar** — the inertial/localizability face of Position ↔ Space — formalized in §5 as `P = √(P_S · P_L)`, where the mass proxy `P_M` is one component of `P_L`, not a peer of `F`, `P`, `A`. The core triad remains `U = ∛(F · P · A)`. *(Proved unique in Part II, §19 / PMD-6, with the elasticity split in §20 / PMD-7 showing the nesting does not double-weight Position.)*

So mass plays **two** roles for Position: it sets the *rest-frame localization scale* (§2) **and** the *inertial response to a change of momentum* (§3) — **not** a "cost of moving" (a free body coasts arbitrarily far at constant momentum with no force; mass resists **acceleration**, not displacement). Space answers "*is the system relationally distinguishable / placeable?*"; mass answers "*what is its rest-frame localization scale and inertial response?*" — the spatial face and the inertial face of one currency.

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4. Mass as *concentrated context* (the interpretive leap — L3)

Why *read* mass as "context"? Because **several** mass contributions and effective inertial parameters are **interaction- or environment-dependent** — *without* this meaning that **all** invariant mass is relational:

- **Mach's principle.** Inertial mass is (on the Machian reading) determined by the distribution of **all other matter** in the universe — inertia is a relation to the cosmic context, not an intrinsic label. *(Status honestly stated: Mach's principle is a real, historically central idea — it motivated Einstein — but it is **not fully established** in GR; frame-dragging realizes it only partially. See §8.)*
- **The Higgs mechanism.** In the Standard Model, elementary fermion and weak-boson masses arise from **coupling to the nonzero Higgs vacuum expectation value** — an *interaction-generated* mass term, **not** friction or drag through a medium. A field without such a coupling gets no Higgs mass term. PMD reads this *interaction origin* of some mass as context — but the Higgs mechanism does not by itself establish that reading, and most of the mass of ordinary matter is **QCD dynamics, not Higgs** (§8).
- **Relational position** *(philosophical, not a physics result)*. A "here" is defined only **against** a context of other things; with no context, "here" has no referent.

Scope (honest). Mach's principle is an *open* hypothesis (only partially realized in GR via frame-dragging); the Higgs mechanism is *specific and established*; relational position is *philosophical*. PMD interprets the **interaction-dependence** of mass as context — it does **not** claim that *all* invariant mass is Machian or relational.

U-Theory reading (speculative, L3): *mass is what a system pays to be **counted as somewhere** by the rest of the universe.* Concentrate context → localizability and inertial response appear together; remove context → the system delocalizes and its "here" dissolves. This is `APPENDIX_MMT` at the Position axis: MMT says **matter = crystallized meaning**; PMD says, more specifically, **mass = crystallized context** — the Position-currency's share of that crystallization. *(Epigraph "begun to weigh" is figurative; the body claim is inertial mass / rest-frame localizability, not gravitational weight.)*

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5. The dual price of Position (the law)

> ⬛ **THE POSITION LOCALIZATION LAW (PMD)**
>
> **A definite location is bought with two non-substitutable sub-prices of the Position currency:**
> **(i) Space — *constrained spatial support / distinguishability*:** become relationally placeable — a constrained spatial support and a context-resolved "where." *(Exclusion / non-overlap is **one** realization, not the universal definition — bosons and fields can share support.)*
> **(ii) Mass — *localizability / inertia*:** in the massive rest-frame sector, set the localization scale (reduced Compton `ƛ_C = ħ/mc` finite) and the **inertial response to a change of momentum** (`F = dp/dt` , i.e. `F ≈ m·a` for `v ≪ c`) — *not* a resistance to

> displacement itself.
> **The zero-mass limit is NOT automatically "delocalized" (v1.2.1 correction).** At `m → 0` the **massive rest-frame localization channel `P_M` vanishes** (`ƛ_C → ∞`; no rest frame; no rest-frame "here"; **zero invariant rest mass**, yet still carrying energy-momentum and gravitating, at the speed of light). But the overall localizability meter `P_L` need **not** be zero — a massless quantum is still scored through the detection channel `P_D` (the massless adapter `g_0`), so it is **not** misclassified as zero-Position. A genuinely *unpaid* Position requires **both** `P_S → 0` **and** `P_L → 0`. **Localization is the crystallization of context into a finite localization scale — carried by mass in the massive rest-frame sector, and by the detection channel for massless quanta.**

face of Position	currency	physics anchor	the zero-limit (unpaid)
be relationally placeable	Space (spatial support / distinguishability)	constrained support; context-resolved placement	no distinguishable "where" — the canon's `P→0`
be localizable & resist Δmomentum	Mass (context/inertia)	`ƛ_C = ℏ/mc`; `F = dp/dt`; Mach; Higgs	`ƛ_C → ∞`: the **mass channel `P_M`** vanishes (no rest-frame "here", no rest inertia; still carries `E`–`p`) — but `P_L` via `P_D` need not vanish

Both faces must be paid for a system to *have a position at all*; a system rich in one and starved of the other has an **ill-defined** Position — which, under `U = ∛(F·P·A)`, drags the whole product down (non-compensation).

Operational placeholder for "context" (upgraded in Part II). In this appendix, **context** is a *working label* for the interaction / environment dependence that makes a system's "here" and inertial response well-defined. The minimal placeholder is:

```
Context_proxy  ≔  { couplings, boundary conditions, reference frame,
                    and co-present degrees of freedom that enter the
                    system's localization or inertial description }
```

This is **not** a claim that context is a new observable, nor that it equals invariant mass. **Part II (§10, §14) upgrades this placeholder to a concrete definition:** context ≔ the **measurement channel `𝓒`** itself (`p(y|x,𝓒)`); its *localization content* is quantified by the **Fisher information `J_𝓒`** about the position parameter `x`, with additivity across independent channels (PMD-3) and a data-processing bound (PMD-4). *(The channel is the object; `J_𝓒` is how much it tells you about **where**.)* Every numerical use of PMD must still declare its own proxy explicitly (§8).

Nesting (keeps Position as ONE pillar — not a fourth price), with a domain gate. The two faces compose *inside* Position, geometrically, so a zero in either zeros Position. The localizability face is a **general localization/detection meter `P_L`**, defined **(v1.4, adopting**

§17) as a weighted geometric blend of a **detection channel** `P_D` and the **mass-capacity proxy** `P_M`, with the **pre-registered default** `η = ½`:

```
POSITION ⇔ SPACE
   ├─ P_S : constrained spatial support / relational distinguishability     ∈
[0,1)
   └─ P_L : localizability / localization-detection meter                   ∈
[0,1)
            ├─ massive rest-frame sector : P_L = P_D^(1−η) · P_M^η ,  η = ½
default   (§17)
            │        P_D = detection/Fisher channel (§15) ;  P_M = g_M(L*/
ƛ_C) (§16)
            │        special case η = 1 → P_L = P_M   (mass-only, legacy
Compton reading)
            └─ massless field sector     : P_L = P_D = g_0( Pr[detect in
R,Δt], σ_R, frame )

   P = P_PMD = √( P_S · P_L )        U = ∛( F · P_PMD · A )
```

This blend discharges the two key negative controls **by construction** (§8, §17): a *massive-but-delocalized* state has `P_D = 0 ⇒ P_L = 0`; a *massless-but-detected* quantum has `P_L = P_D > 0` (never zero-Position). Mass is **not** a fourth top-level pillar; it is **one component** (via `P_M`) of the nested `P_L` meter, and the mass-only `P_L = P_M` reading survives as the `η = 1` special case. `g_M`, `g_0`, `η`, the reference scale `L*`, the normalization, and a missing-data rule are **declared before use** and are **calibration candidates, not canon** (§8). *(Part II §16–§17 derives the admissible forms and the adapter;* `η` *must be pre-registered, never fit post hoc.)*

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6. The conjugacy — position, momentum, and the massless limit

A **massless** excitation (e.g. a photon) has **no rest frame** and **no massive-particle rest-frame localization scale** — it is not a little ball with a coordinate. But it is **not** "everywhere": a single-photon **wave packet** can be prepared with finite spatial–temporal localization and produce **local detection events** (there is simply no sharp Newton–Wigner one-particle position observable for it). So masslessness removes the *rest-frame particle-"here"*, **not all localization.** Acquiring mass gives a massive particle a rest-frame localization scale (`ƛ_C`) and an inertial response — it does **not** follow that massless quanta cannot be localized at all.

This mirrors the **position–momentum uncertainty** `Δx·Δp ≥ ħ/2`: a sharp "here" (`Δx` small) costs a large momentum spread. **The conjugate of position is momentum, not mass** — but mass enters the trade through the relativistic dispersion `E² = p²c² + m²c⁴` and the localization scale `ƛ_C = ħ/mc`. (In general `p = γ·m·v`; a photon carries `p = E/c` despite **zero** rest mass — so `p = m·v` is only the `v ≪ c` approximation and is never applied here to massless or relativistic cases.) Sharpening a "here" costs momentum spread; for a **massive** particle **mass sets the rest-frame scale** of that trade. But it is **momentum, not mass**, that is conjugate to position — and a massless quantum can still be wave-packet-localized. The defensible statement is **not** "delocalized ↔ massless"; it is only that mass **sets a massive particle's rest-frame localization scale.**

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7. Binding it to the whole corpus

- `SSS` **/** `U = ∛(F·P·A)` **:** the **Position pillar** now carries an explicit *dual meter* — spatial support / distinguishability **and** localizability (with mass as one proxy). A system with strong Form and Action but near-zero relational placement / localizability has a near-zero Position, so `U` collapses. This is the **physical face** of the corpus's central thesis (autobiography, POS): *strong Form and Action do not compensate a Position near zero. Illustrative flavour only (not a classification):* a system rich in Form and Action but poor in relational placement / localizability scores low on Position and, by non-compensation, low on `U`. **This must not be read literally for a photon** — a massless quantum is locally detectable and is *not* zero-Position (§5 adapter `g_0`, §8).
- `MMT` **:** matter = crystallized meaning; PMD refines the Position axis — **mass = crystallized context** *(interpretive)*. The Big Bang as maximal concentration (`max 𝓜`) is, *under the Machian reading* (§4, §8 — an open question, not settled physics), context concentrating until inertial response appears; without that reading, the same event is simply maximal energy density, and PMD adds no extra claim.
- `DIM` **:** required meaning scales with dimension; localization is a *spatial-dimensional* act, and PMD says the massive rest-frame sector of that act is paid partly in mass/context.
- `QMC` **(measurement):** a **localized position measurement records a localized outcome** and, under standard state-update modelling, **conditionally prepares a more localized post-measurement state** — via an interaction (context) with an apparatus. PMD reads that interaction as paying the context price, without committing to any one interpretation of measurement. *(Formalized in Part II §11–§15: the apparatus channel contributes Fisher information* `J_𝒞` *about position.)*
- `POS` **(Principle of Sequence):** `F → P → A`. Physically, a system needs **Form** (persistence in time) and then a **Position** (a "here," bought with space **and**, in the massive sector, mass/context) *before* it can spend **Energy** in Action. No "here" → no base to act from — the Forbidden Action at the level of physics.

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8. Falsifiability and honest scope

What is rigorous (not speculative):

- `ƛ_C = ħ/mc` (reduced) as the **one-particle, rest-frame** localization scale; `m → 0 ⇒` no rest-frame localization scale, `m ↑ ⇒` a finer one — standard relativistic QM.
- Inertia as resistance to a *change of momentum* (`F = dp/dt`; `F ≈ m·a` only for `v ≪ c`) — standard mechanics.
- Mass from Higgs coupling — Standard Model. **Caveat:** the Higgs gives **elementary** rest masses; **most** of the mass of ordinary matter (protons, neutrons) is **QCD binding / field energy**, not a Yukawa coupling. PMD reads both as *context becoming inertial* — but only

the Higgs part is a *direct* SM mass-generation mechanism. *Why the composite (QCD) case strengthens the reading rather than weakening it:* nucleon mass is not carried by its constituents in isolation — it is the **energy of the internal interactions** (gluon fields, confined quark motion) of the system *with itself*. A proton's inertia is therefore the inertial face of its own **internal relational context** — the "mass = crystallized context" reading applied to a system's interactions with its own parts, exactly the composite it is meant to describe. The Higgs case is *interaction with an external field* (the VEV); the QCD case is *interaction of the system with itself* — both are context becoming inertial, and the second is the more direct instance. *(Avoid "weight" language here: the claim is about inertial mass / rest energy, not gravitational weight. The signed mass-defect subtlety — QCD adds mass, nuclear/atomic binding subtracts it — is formalized in Part II §22 / PMD-8 as `χ_int`.)*

What is speculative interpretation (L3/L4), and its honest hooks:

- **"Mass = concentrated context / the Position-currency's inertia face"** is a *reading*, not a derivation. It does not add to or modify GR/QFT; it re-describes them in F/P/A terms.
- **The strong converse — "no location ⇒ no mass" — is NOT standard physics and is flagged as the most speculative claim.** A *delocalized but massive* state exists in ordinary QM (a momentum eigenstate of a massive particle is spread over all space yet has mass). PMD's honest claim is the *rigorous direction only*: **masslessness forbids the rest-frame particle-"here"** (no rest frame; no sharp one-particle position operator for the photon), **while mass enables a finite localization scale**; local detection of massless quanta still occurs. The reverse (delocalized ⇒ massless) is offered as a suggestive lens, not asserted.
- **Mach's principle is genuinely open.** If inertia were shown to be entirely independent of the cosmic matter distribution, the "context confers mass" reading weakens to a metaphor. If frame-dragging / relational-inertia results strengthen the Machian picture, it strengthens. PMD is therefore **hostage to a real open question in physics**, and says so.

Null model, empirical content, and required controls (RH).

- **Present empirical content (honest):** as of v1.3, PMD makes **no observation that standard ƛ_C, relativistic kinematics, QFT, and GR do not already make**. The functions `g_M`, `g_0`, and the scale `L*` are calibration candidates, not fitted laws. This is not a soft admission — Part II **proves** it: the **no-go theorem (§23 / PMD-9)** shows that if the PMD context variable is any deterministic function of the standard variables, `C = f(X_std)`, then `I(Y ; C | X_std) = 0` for every observable `Y`. **PMD earns empirical content only against an independently measured context observable `C_R` with `I(Y ; C_R | X_std) > 0`.**
- **Null model:** `H₀: every observation is explained by the standard ƛ_C, pᵘ, and QFT/GR, with no PMD context variable.`
- **What would count as content:** a mass/localization meter that is (i) independently measurable, (ii) adds predictive power over `H₀`, and (iii) never misclassifies massless, locally-detected systems.
- **Required negative controls before any use: (a) massive-but-delocalized** (a momentum eigenstate — massive yet spread out → refutes "delocalized ⇒ massless"); **(b) massless-but-locally-detected** (a photon wave packet → refutes `P_L = 0` for

massless); **(c) inertial-motion null** (constant-velocity coasting, no force → refutes "mass resists displacement"); **(d)** `P_S` **-only rival** (must the mass sub-meter beat a Position score built from spatial support alone?). *(These are discharged by construction in the upgraded adapter, Part II §17.)*

Epistemic level (Part I): L3/L4 speculative — a coherent, corpus-consistent *interpretation* resting on a rigorous physical core. It makes no claim of empirical proof and adds no new governing equation. It sharpens the meaning of the Position currency; it does not re-derive physics. *(The formal core that IS provable — the localization–energy bound, Fisher additivity, the nested-mean uniqueness, and the no-go — is Part II.)*

════════════════════════════ PART II — FORMAL LAYER (PMD-MATH) ════════════════════════════

> **From philosophical analogy to a formal measurement framework** — what can be *proved*, what is a *modelling choice*, and what is *forbidden* without a new observable.
>
> **One-line thesis (the strongest defensible claim).** In the massive non-relativistic sector, **invariant mass upper-bounds the spatial Fisher information extractable per unit kinetic energy** (`F_Q/K ≤ 8m/ħ²`); the **measurement context** determines what fraction of that capacity is actually realized (`J_𝒞 ≤ F_Q`). Therefore **mass sets capacity; context realizes accessibility** — two distinct, complementary roles, *not* an identity.

9. Consistency audit

9.1 What CAN be proved mathematically (Part II)

- the reciprocal identity `m·ƛ_C = ħ/c` (§11 / PMD-1);
- the mass–energy–localization inequality `F_Q ≤ 8mK/ħ²` (§12 / PMD-2);
- additivity of context Fisher information across independent channels (§13 / PMD-3);
- the data-processing (monotonicity) bound: losing context cannot raise position information (§14 / PMD-4);
- the admissible one-parameter family for `g_M` from a log-odds axiom (§16 / PMD-5);
- uniqueness of the nested geometric mean under separability + symmetry + idempotence (§19 / PMD-6);
- the elasticity split showing the nesting does **not** double-weight Position (§20 / PMD-7);
- the exact participation of interactions in composite invariant mass (§22 / PMD-8);
- the condition for empirical surplus over the null model, as a **no-go** (§23 / PMD-9).

9.2 What CANNOT be proved by mathematics alone

The following stay **interpretive (L3/L4)** and are *not* elevated by Part II:

```
mass = context
localization = crystallization of context
cosmic context causes all inertia
```

Mathematics can derive consequences *once a formal definition of "context" is accepted*, but it cannot prove that definition corresponds to a fundamental physical reality.

9.3 Residual wording defect in Part I (fixed in v1.2.1)

Part I §5's law box once carried "the unpaid limit is the massless, **delocalized** state," while §6/§8 correctly reject "massless ⇔ delocalized." The mathematically correct statement — now enforced in §5 — is: at `m → 0` the **massive rest-frame channel** `P_M` **vanishes**, but the overall localizability meter `P_L` need **not** be zero (it is carried by the detection/Fisher channel `P_D`). Masslessness removes the rest-frame particle-"here", not all localization.

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10. Formal domain

Let:

```
L*  > 0     a pre-registered, task-specific localization scale        [m]
m   ≥ 0     invariant mass                                            [kg]
ƛ_C = ħ/(mc)   reduced Compton wavelength (for m>0)                   [m]
x  ∈ ℝ^d    spatial-translation parameter (the "where")
ρ_x         quantum state after translation by x
𝒞           the measurement context
P_S,P_D,P_M,P_L,P ∈ [0,1]   (P_D, P_S, P_M lie strictly in [0,1) — they
saturate toward 1, never reach it)
```

Spatial translation is generated by momentum `p̂`:

```
ρ_x = e^(−i x p̂ / ħ) · ρ_0 · e^(+i x p̂ / ħ)
```

The context $\mathcal{C}$ is represented by a POVM / classical measurement channel:

```
p(y | x, 𝒞) = Tr( ρ_x · E_y^(𝒞) )
```

so **"context" stops being a word.** It is:

```
𝒞 = { coupling, boundary, frame, measurement channel }
```

— and it fixes *how much information about x can actually be extracted.*

11. Theorem PMD-1 — Compton reciprocity · [L1: identity]

For $m > 0$: $ƛ_C = \hbar/(mc)$, hence

$$\blacksquare \quad m \cdot ƛ_C = \hbar / c \qquad \text{(mass × localization scale is a universal constant)}$$

Proof. $m \cdot ƛ_C = m \cdot \hbar/(mc) = \hbar/c$. ■ Also $dƛ_C/dm = -\hbar/(cm^2) < 0$, so the Compton scale is strictly decreasing in mass.

Non-relativistic inertial corollary. At fixed force F and $v \ll c$, $a = F/m$, so

$$ƛ_C / a = (\hbar/mc) / (F/m) = \hbar / (cF) \quad \Rightarrow \quad \blacksquare \; ƛ_C \propto a \propto 1/m \quad \text{(at fixed F)}$$

Reading (not a proof of "mass = context"). The *same* parameter m that shrinks the massive rest-frame localization scale also shrinks the acceleration response at fixed force. The two PMD anchors — localization scale (§2) and inertia (§3) — are governed by **one** physical quantity. That is a genuine structural unification; it is *not* a claim that mass is context.

12. Theorem PMD-2 — Localization Information–Energy bound · [L2: proved, non-relativistic]

(The strongest new formal result — the defensible quantitative core of PMD.)

For the translation family ρ_x generated by $\hat{p}$, the **quantum Fisher information** for estimating x obeys

$$F_Q(\rho_x) \le 4 (\Delta p)^2 / \hbar^2 \qquad \text{(equality for pure states)}$$

With non-relativistic kinetic energy $K = \langle p^2\rangle / (2m)$ and $(\Delta p)^2 = \langle p^2\rangle - \langle p\rangle^2 \le \langle p^2\rangle = 2mK$:

$$\blacksquare \quad F_Q \le 8 m K / \hbar^2 \qquad \text{(equality for a pure, zero-mean-momentum state)}$$

Proof. Combine $F_Q \le 4(\Delta p)^2/\hbar^2$ (QFI of a unitary family = 4·Var of its generator, with $\le$ for mixed states) and $(\Delta p)^2 \le 2mK$. ■

Cramér–Rao corollary. For N independent measurements, $\mathrm{Var}(\hat{x}) \ge 1/(N \cdot F_Q)$, hence

$$\blacksquare \quad \mathrm{Var}(\hat{x}) \ge \hbar^2 / (8 N m K)$$

Capacity reading (defensible). Per unit kinetic-energy budget, larger mass permits a lower fundamental error floor on estimating a spatial displacement:

```
■  F_Q / K ≤ 8 m / ħ²        ("mass = localization-information leverage per unit energy")
```

This is far more defensible than "localization is bought with mass": it is a *proved inequality*, explicitly scoped to the non-relativistic massive sector, that assigns mass a precise operational role (capacity), not a metaphysical one. **Scope caveat:** the bound is **non-relativistic** — it uses $K = \langle p^2\rangle/2m$, and makes **no claim** in the relativistic regime governed by $E^2 = p^2c^2 + m^2c^4$ (there $K \neq \langle p^2\rangle/2m$ and the $8mK/\hbar^2$ form does not apply).

---

13. Theorem PMD-3 — context Fisher information is additive · [L2: proved]

For n conditionally independent channels, $p(y_1,\dots,y_n \mid x) = \prod_k p_k(y_k \mid x)$, with per-channel classical Fisher information $J_k(x) = \mathbb{E}[(\partial_x \ln p_k)^2]$:

```
■  J_𝒞(x) = Σₖ Jₖ(x)
```

Proof. The score of the product is the sum of scores, $s = \Sigma_k s_k$. Then $J\_\mathcal{C} = \mathbb{E}[(\Sigma s_k)^2] = \Sigma_k \mathbb{E}[s_k^2] + 2 \Sigma_{i<j} \mathbb{E}[s_i s_j]$. For regular models $\mathbb{E}[s_k] = 0$; under conditional independence $\mathbb{E}[s_i s_j] = \mathbb{E}[s_i]\mathbb{E}[s_j] = 0$. Hence $J\_\mathcal{C} = \Sigma_k J_k$. ■

Interpretation. This gives a precise meaning to "more context": *more independent relational channels ⇒ more position information.* It does **not** say context creates mass — only that relational context increases the **measurability** of "where."

---

14. Theorem PMD-4 — losing context cannot raise localization information · [L2: proved]

For a coarse-graining Markov chain $x \to Y \to Z$:

```
■  J_Z(x) ≤ J_Y(x)        (Fisher-information data-processing inequality)
```

Proof (sketch, under standard regularity — support of Y independent of x, and $\partial/\partial x$ interchangeable with integration**).** The coarse-grained score is a conditional expectation, $s\_Z = \mathbb{E}[s\_Y \mid Z]$; by Jensen, $\mathbb{E}[s\_Z^2] = \mathbb{E}[\ \mathbb{E}[s\_Y|Z]^2\ ] \leq \mathbb{E}[s\_Y^2]$, i.e. $J\_Z \leq J\_Y$. ■

The defensible context thesis. Removing or coarse-graining relational channels **cannot increase** the information with which a system is localizable. (The converse — that adding context *creates* location rather than reveals it — is *not* claimed.)

---

15. Operational localization (detection) meter P_D · [L3: modelling choice]

Define the dimensionless Fisher ratio $z_D = L^{*2} \cdot J_{\mathcal{C}}$ and

$$\blacksquare \quad P_D = z_D / (1 + z_D) = L^{*2} \cdot J_{\mathcal{C}} / (1 + L^{*2} \cdot J_{\mathcal{C}})$$

Properties (all verified): $0 \le P_D < 1$; $P_D(0) = 0$; $P_D \to 1$ as $J \to \infty$; $\partial P_D/\partial J = L^{*2}/(1+L^{*2}J)^2 > 0$. It is dimensionless, bounded, strictly monotone, saturating, and **valid for both massive and massless local detections** — a cleaner realization of Part I's massless adapter g_0.

---

16. Theorem PMD-5 — the admissible family for g_M · [L3: conditional derivation]

Let $x_M = L^*/\lambda_C = mcL^*/\hbar = m/m^*$, with reference mass $m^* = \hbar/(cL^*)$. Require $g_M : (0,\infty) \to (0,1)$ continuous, strictly increasing, $g_M(1) = ½$, and **log-odds additive under multiplicative rescaling** of x_M:

$$\ell(x \cdot y) = \ell(x) + \ell(y), \quad \text{where} \quad \ell(x) = \ln[\, g_M(x) / (1 - g_M(x)) \,]$$

Then the continuous solutions of that Cauchy equation are $\ell(x) = \alpha \cdot \ln x$ ($\alpha > 0$), so

$$\blacksquare \quad g_M(x) = x^\alpha / (1 + x^\alpha)$$

Proof. $\ell(xy)=\ell(x)+\ell(y)$ continuous ⇒ $\ell(x)=\alpha \ln x$; exponentiating $g/(1-g)=x^\alpha$ and solving gives the logistic form. ■

Preferred Fisher exponent. If the mass channel is read as an inverse-variance capacity $J_M \propto 1/\lambda_C^2$, the natural exponent is $\alpha = 2$, giving

$$\blacksquare \quad P_M = (L^*/\lambda_C)^2 / (1 + (L^*/\lambda_C)^2) = (m/m^*)^2 / (1 + (m/m^*)^2)$$

This is **not** a new physical law — it is a canonical normalization candidate justified by dimensional analysis, inverse-variance reading, boundedness, and scale symmetry. $\alpha \in \{1, 2, \text{free}\}$ is a calibration/model-comparison question, not settled.

---

17. Upgraded massive adapter (Part I §5 `P_L = P_M` → v1.3+ proposal) · [L3]

Part I's current `P_L = P_M` in the massive sector cannot distinguish a **localized** massive particle from a **massive momentum eigenstate delocalized over all space**. The proposed fix folds in the detection channel:

```
■  massive sector:   P_L = P_D^(1−η) · P_M^η ,   0 ≤ η ≤ 1
■  massless sector:  P_L = P_D
```

At the symmetric point `η = ½`, `P_L = √(P_D · P_M)`. The two essential negative controls (Part I §8) are then satisfied **by construction**:

```
massive-but-delocalized :  P_D = 0, P_M > 0  ⇒  P_L = 0     (correctly not
localized)
massless-but-detected   :  m = 0,    P_D > 0  ⇒  P_L = P_D>0 (correctly not
zero-Position)
```

This is strictly stronger than the `P_L = P_M` branch because it embeds the two controls into the formula rather than relying on a case split. **Status:** recommended for adoption into the Part I §5 core meter at **v1.3+**; `η` must be **pre-registered**, never fit post hoc.

---

18. Spatial-support meter `P_S` · [L3: modelling choice]

Let `ρ(x)` be a normalized spatial distribution over a pre-declared reference region `Ω`, and `u_Ω = 1/|Ω|` the uniform. Define the distinguishability information as a KL contrast:

```
D_S = D_KL(ρ ‖ u_Ω) = ∫_Ω ρ(x) · ln[ ρ(x) / u_Ω(x) ] dx ≥ 0   (Gibbs; =0 iff
ρ=u_Ω a.e.)
```

and

```
■  P_S = 1 − e^(−D_S / D*) ,   D* > 0
```

so `P_S ∈ [0,1)`, `P_S = 0 ⇔ ρ = u_Ω` (no distinguishable "where"), `∂P_S/∂D_S = (1/D*)e^(−D_S/D*) > 0`. This operationalizes "constrained support" as an information contrast against a chosen context.

---

19. Theorem PMD-6 — uniqueness of the nested geometric mean · [L3: conditional]

Assume the Position aggregator is **multiplicatively separable**, `G(P_S,P_L) = f(P_S)·f(P_L)`, and require symmetry, idempotence `G(t,t)=t`, zero-annihilation, and monotonicity. Idempotence gives `f(t)² = t`, so `f(t) = √t`, hence

```
■  G(P_S, P_L) = √(P_S · P_L)
```

The geometric nesting is therefore **not arbitrary** — it is forced once separability + symmetry + idempotence + non-compensation are accepted. ■

Scope of the uniqueness (honest). This is uniqueness *within the axiom set*. Rival aggregators that violate one axiom — the minimum `min(P_S,P_L)`, a soft-min, or a weighted geometric mean `P_S^a · P_L^b` with `a ≠ b` (breaks symmetry) — are **not refuted**; they simply fall outside {multiplicative separability, symmetry, idempotence}. CEPT `CORE.PMD.08` records this as *conditionally proved*, and §24 lists rival aggregators as an explicit model-comparison arm.

---

20. Theorem PMD-7 — full nested U and elasticity (no double-weighting) · [L2 given the meters]

With `P = √(P_S·P_L)` and `U = ∛(F·P·A)`:

```
■  U = F^(1/3) · A^(1/3) · P_S^(1/6) · P_L^(1/6)
```

Log-elasticities:

```
∂lnU/∂lnF = 1/3      ∂lnU/∂lnA   = 1/3
∂lnU/∂lnP_S = 1/6    ∂lnU/∂lnP_L = 1/6
```

The two Position sub-meters together carry `1/6 + 1/6 = 1/3` — **exactly** Position's original share. The nesting **splits** Position's existing weight into two equal halves; it does **not** add weight. *(With the §17 adapter `P_L = P_D^(1−η)·P_M^η`, the `P_L` share `1/6` splits further into `P_D:(1−η)/6` and `P_M:η/6` — at `η=½`, `1/12` each; see §21.)* This is what keeps PMD compatible with `SSS`, where the three top-level pillars aggregate geometrically.

---

21. Full proposed PMD operational formula

Massive sector (`η` -weighted adapter):

```
P_M = (mcL*/ħ)^α / (1 + (mcL*/ħ)^α)
P_D = L*²·J_𝒞 / (1 + L*²·J_𝒞)
P_L = P_D^(1−η) · P_M^η
P   = √(P_S · P_L)

■  U_PMD = F^(1/3) · A^(1/3) · P_S^(1/6) · P_D^((1−η)/6) · P_M^(η/6)
```

Massless sector (`P_L = P_D`; there is **no** `P_M = 0`, only an *inapplicable* mass channel):

```
■  U_PMD^(0) = F^(1/3) · A^(1/3) · P_S^(1/6) · P_D^(1/6)
```

---

22. Theorem PMD-8 — interactions participate in composite invariant mass · **[L1: standard, with signed correction]**

For a closed system with total four-momentum `P^μ`:

```
■  M²c² = P_μ P^μ ,   and in the centre-of-momentum frame (P=0):   ■ Mc² = E_COM
```

Writing `H = Σᵢ mᵢc² + H_kin,int + H_field + H_int`:

```
■  M − Σᵢ mᵢ = ( ⟨H_kin,int⟩ + ⟨H_field⟩ + ⟨H_int⟩ ) / c²
```

This is the exact, rigorous sense in which a system's **internal relations participate in its invariant mass**.

Essential sign correction (honest). The sign is **not** universally positive:

- **Nucleons:** QCD dynamics contribute a *large positive* mass over the sum of current-quark masses.
- **Atomic / nuclear bound states:** binding energy gives a *mass defect* (mass **below** the sum of separated parts).

So the defensible statement is `interaction context MODIFIES invariant mass`, **not** `more context always means more mass`. Define a **signed** interaction-mass fraction:

```
χ_int = (M − Σᵢ mᵢ) / M
   χ_int > 0 : interactions raise total mass over the constituent baseline (e.g. QCD)
   χ_int < 0 : binding mass defect (e.g. nuclei, atoms)
   χ_int = 0 : no net interaction contribution vs baseline
```

`χ_int` is a real candidate for a *partial* operationalization of "internal relational context."

Why the composite (QCD) case still strengthens the reading (see Part I §8): a nucleon's mass is the energy of its **own internal interactions with itself** made inertial — the "mass = crystallized context" reading applied to a system's relations with its own parts. The signed correction simply forbids the *slogan* "more context ⇒ more mass" while keeping the *structural* claim "interaction context is inertial."

---

23. Theorem PMD-9 — no-go for automatic empirical surplus · **[L1: information theory — the honesty gate]**

Let the standard model of the phenomenon use all standard variables `X_std = (m, p^μ, ρ, H_int, 𝒞_measurement, …)`, and let the PMD "context" variable be a **deterministic function** of them, `C = f(X_std)`. Then for **every** observable `Y`:

```
■  I(Y ; C | X_std) = 0
```

Proof. `C = f(X_std) ⇒ H(C | X_std) = 0`, so `I(Y;C|X_std) = H(C|X_std) − H(C|Y,X_std) = 0 − 0 = 0`. ■

Consequence (the discipline). If "context" is merely a **relabelling** of variables standard physics already uses, PMD **cannot** add predictive information. Empirical surplus **requires** an *independently measured* context observable `C_R` with

```
■  I(Y ; C_R | X_std) > 0
```

This is the formal version of the Part I §8 null model `H₀`. PMD is honest precisely because it proves its own emptiness absent a new observable.

---

24. Pre-registered PMD empirical model (how surplus would be earned)

Outcome must be a *raw* observable, never the constructed `P_D`. Using `P_D` as the target would be **circular** — it is built from the same channel `𝒞` as the predictors. Let `Y` be a directly measured quantity: the empirical localization error `Var(x̂)`, a detection hit-rate in region `R`, or a resolved position residual.

Null model. `H₀:  g(Y) = f_std(m, Δp, E, POVM, geometry) + ε`

PMD model.

```
H₁:  g(Y) = β₀ + β_m·ln x_M + β_C·C_R + β_{mC}·(ln x_M)·C_R + ε
```

where `g` is a fixed link (e.g. `ln Var(x̂)`) and `C_R` is an **independently measured** relational-context variable (**not** a deterministic function of `X_std` — otherwise PMD-9 forces zero surplus).

PMD earns content only if, on held-out data, `β_C ≠ 0` or `β_{mC} ≠ 0`, and

```
Δ_CV = Loss(H₀) − Loss(H₁) > 0     (pre-set statistical margin),
equivalently  I(Y; C_R | X_std) > 0.
```

`RH` requires exactly this contest against the null and against rivals — not mere internal mathematical coherence.

---

25. Worked example for `P_M` (illustration only)

With `α = 2`, `L* = 1 fm`, and `P_M = (L*/ƛ_C)² / (1 + (L*/ƛ_C)²)`:

particle	ƛ_C	L*/ƛ_C	P_M
electron	386.159 fm	0.00259	6.7×10^{-6}
proton	0.2103 fm	4.755	0.958
photon (massless)	— (`ƛ_C→∞`)	—	**N/A** — mass channel inapplicable; scored via `P_D > 0` (§17 massless adapter), **not** `P_L = 0`

This does **not** mean the proton "has Position 0.958" and the electron "has almost none." It means only: *relative to the task scale `L* = 1 fm`, the proton's mass Compton-capacity proxy is near-saturated and the electron's is not.* Choose a different `L*` and the numbers change. Therefore `L*` must be **task-specific, pre-registered, and never chosen after seeing the result.** *(All numbers verified numerically.)*

---

26. Claim–Evidence–Prediction–Test map

ID	Claim	Status	Basis	Test
CORE.PMD.01	`m·ƛ_C = ħ/c`	established	identity	unit check
CORE.PMD.02	mass ↔ acceleration-susceptibility reciprocal	established (NR)	`a=F/m`	fixed-force test
CORE.PMD.03	`F_Q ≤ 8mK/ħ²`	proved (scoped)	QFI + kinetic energy	displacement estimation
CORE.PMD.04	context channels add Fisher info	proved	conditional independence	multi-detector experiment
CORE.PMD.05	coarse-graining lowers position info	proved	data processing	sensor removal
CORE.PMD.06	`g_M = x^α/(1+x^α)`	conditionally proved	log-odds axiom	calibration
CORE.PMD.07	`α = 2`	model choice	inverse-variance scaling	model comparison
CORE.PMD.08	`P = √(P_S·P_L)`	conditionally proved	separability + symmetry	rival aggregators
CORE.PMD.09	interactions modify composite mass	established	invariant mass	mass-defect data
CORE.PMD.10	context *always* increases mass	**false in general**	sign varies (χ_int)	binding controls
CORE.PMD.11	mass = context	L3 interpretation	no derivation	independent `C_R`
CORE.PMD.12	PMD adds prediction	**not yet**	requires `I>0`	held-out test

27. Variable & equation registry

symbol	meaning	units
`m`	invariant mass	kg
`ƛ_C`	reduced Compton wavelength	m
`L*`	target localization scale	m
`m* = ħ/(cL*)`	reference mass	kg
`x_M = m/m*`	Compton ratio	—
`J_𝒞`	classical Fisher information for position	m^{-2}
`F_Q`	quantum Fisher information	m^{-2}
`D_S`	KL support information	—
`P_S`	support / distinguishability score	[0,1)
`P_D`	detection / localization score	[0,1)
`P_M`	massive Compton-capacity score	[0,1)
`P_L`	combined localizability score	[0,1]
`η`	mass-proxy weight	—
`χ_int`	signed interaction-mass fraction	—
`C_R`	independently measured context	protocol-specific
`U`	nested stability score	[0,1]

28. Dependency graph

```
Standard physics
   ├── ƛ_C = ħ/(mc)
   ├── F = dp/dt
   ├── QFI / Cramér–Rao
   └── M²c² = P_μ P^μ
        │
        ▼
PMD mathematical bridge
   ├── mass capacity      P_M
   ├── detection info     P_D
   ├── support info       P_S
   └── interaction frac.  χ_int
        │
        ▼
Nested Position
   P_L = P_D^(1−η) · P_M^η
   P   = √(P_S · P_L)
        │
        ▼
SSS:  U = ∛(F · P · A)
        │
        ▼
Empirical test:  H₀ vs H₁ ,  I(Y ; C_R | X_std) > 0 ?
```

---

29. Open issues / calibration tasks (v1.5 roadmap)

1. Corrected §5 wording (done — v1.2.1).
2. ~~Replace `P_L = P_M` with the `P_D` -inclusive adapter~~ — **done in v1.4** (§5 core now `P_L = P_D^(1−η)·P_M^η`, η=½ default; `P_L=P_M` kept as η=1 case).
3. Choose or compare `α ∈ {1, 2, free}`.
4. Fix `η` by pre-registration, never post hoc.
5. Declare the reference region `Ω` for `P_S`.
6. Set `L*` from task resolution, not from the result.
7. Separate internal-context from external/Machian context.
8. Use the **signed** `χ_int` (interactions can lower mass).
9. Compare four models: `P_S` ; `P_S·P_D` ; `P_S·P_M` ; `P_S·P_D·P_M`.
10. Count PMD as empirically upgraded **only** if `I(Y; C_R | X_std) > 0`.

---

30. Strongest formulation of the hypothesis (replaces "mass = concentrated context" as the core)

> **PMD Localization-Leverage Hypothesis.** In the massive non-relativistic sector, invariant mass bounds the spatial Fisher information available per unit kinetic-energy budget,
>
> ```
> F_Q / K ≤ 8m / ħ²
> ```
>
> and the measurement context bounds how much of that available information is actually extracted,
>
> ```
> J_𝒞 ≤ F_Q .
> ```
>
> Hence mass and context have distinct, complementary roles:
>
> ```
> ■  mass sets capacity ;  context realizes accessibility .
> ```

So the defensible corpus statement is **not** `Mass = Context` but

```
■  Position quality = f( massive capacity , spatial state , measurement context )
```

References & provenance

- **Parent record:** U-Theory / U-Model — DOI **10.17605/OSF.IO/74XGR** · https://u-model.org
- **Sibling appendices:** `MMT` (matter = crystallized meaning), `DIM`, `ST`/DPR (currency registry), `QMC` (measurement/localization), `GEN`, `SSS`, `RH`, `POS`.
- **Physics anchors (established):** the **reduced** Compton wavelength `ƛ_C = ħ/mc` (ordinary `λ_C = h/mc`) & the pair-creation one-particle localization limit; the relativistic dispersion `E² = p²c² + m²c⁴` and photon `p = E/c`; inertia via `F = dp/dt` (`F ≈ ma` at low `v`); the absence of a sharp Newton–Wigner position operator for massless spin > ½; quantum Fisher information & the Cramér–Rao bound; Fisher-information additivity and the data-processing inequality; invariant mass `M²c² = P_μ P^μ`; Mach's principle (historical, only partially realized in GR via frame-dragging); the Higgs mechanism (elementary masses; composite mass mostly QCD binding); the position–momentum uncertainty relation.
- **Author:** Petar Nikolov (ORCID 0009-0001-8669-2276).

Changelog

version	date	change
v1.4	2026-07-25	**Review-hardening (P0/P1/P2).** Adopted the §17 adapter into the §5 core meter: `P_L = P_D^(1−η)·P_M^η` (pre-registered `η=½`; `P_L=P_M` retained as the `η=1` legacy Compton reading) — the two negative controls now hold *by construction*, resolving the §5↔§17 dual-state. Clarified context ≔ **channel** `𝒞` vs its Fisher **content** `J_𝒞` (§5). Added: PMD-2 non-relativistic **scope caveat** (§12); PMD-4 **regularity** note (§14); PMD-6 **rival-aggregator** scope (§19); `[0,1)` ranges (§10, §27); **raw-observable** protocol to kill circularity in §24 (`Y` must not be `P_D`); a massless-photon **negative-control row** (§25); PMD-7 adapter elasticity split (§20); epigraph *figurative* footnote; minimal reading-path guide; §29 renamed to v1.5 roadmap. **No new physics; PMD-2/PMD-9/χ_int/§30 untouched.**
v1.3	2026-07-25	**Single-document merge.** Folded the entire formal layer (former `APPENDIX_PMD_MATH`) into this file as **PART II (§9–§30)**: all theorems PMD-1…9 with proofs, the operational meters (`P_D`, `P_M`, `P_S`), the upgraded adapter, the composite-mass `χ_int`, the empirical no-go, the CEPT map, registry, and dependency graph — nothing dropped. Part I (§0–§8) is the interpretation; per-result epistemic labels live in Part II. Internal cross-references rewired to the merged numbering.
v1.2.1	2026-07-25	Consultant v1.2 + PMD-MATH §2.3 fix: the residual §5 "massless⇒delocalized" wording corrected (at `m→0` the mass channel `P_M` vanishes but `P_L` via `P_D` need not; unpaid Position needs **both** `P_S→0` **and** `P_L→0`).
v1.2	2026-07-25	Consultant review-hardening: anti-fourth-pillar note (§3); operational `Context_proxy` placeholder (§5); `resist Δmomentum` table wording; QCD internal-interaction justification (§8); Machian scope qualifier (§7); explicit **no present empirical surplus** clause; changelog table.
v1.1.2	2026-07-25	Review polish draft: QCD strengthens-the-reading argument; §3 nested-sub-price note; §7 Machian qualifier.
v1.1.1	2026-07-19	§8 discipline through the body: massless ≠ delocalized; inertia resists Δmomentum not displacement; Higgs = VEV coupling not drag; massless adapter `g_0`; `H₀` and negative controls.
v1.1	2026-07-19	Physics-hardened core: reduced Compton `ƛ_C`; relativistic `F = dp/dt`, `p = γmv`, photon `p = E/c`; nested meter `P = √(P_S·P_L)`; mass `P_M` as one proxy only in the massive rest-frame sector.

*Status: **v1.4 single document** — Part I is L3/L4 speculative interpretation on a rigorous physical core; Part II is the formal layer (L1/L2 proved core: `m·ƛ_C = ħ/c`; `F_Q ≤ 8mK/ħ²`; Fisher additivity `J_𝒞 = Σ Jₖ`; data-processing monotonicity; nested-mean uniqueness `P = √(P_S·P_L)`; composite mass `Mc² = E_COM` with signed `χ_int`; empirical no-go `C = f(X_std) ⇒ I(Y;C|X_std) = 0`; L3 modelling choices: `P_M`, `P_D`, `P_S`, `α = 2`, `η`). The one-particle reduced-Compton localization scale is textbook physics; reading mass as the*

*Position-currency's context/inertia price is a U-Theory interpretation, not a derivation; "mass = context" is **not** upgraded by the formal layer and remains L3/L4. Position pays twice — in space, and in mass — **for a massive particle in its rest frame.***